

LAB 7



COMSATS University Islamabad

Programming Fundamentals

Fall 2025

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QUESTION 1: Write a program that prompts the user to enter the latitude and longitude of two points on the earth in degrees and displays its great circle distance. The average earth radius is 6,371.01 km. Note that you need to convert the degrees into radians using the `Math.toRadians` method since the Java trigonometric methods use radians. The latitude and longitude degrees in the formula are for north and west. Use negative to indicate south and east degrees.

LAB 7

INPUT(Q1):

Q1 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q1
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        final double radius = 6371.01;
        System.out.print("Enter point 1 (latitude and longitude) in degrees: ");
        double x1 = input.nextDouble();
        double y1 = input.nextDouble();
        System.out.print("Enter point 2 (latitude and longitude) in degrees: ");
        double x2 = input.nextDouble();
        double y2 = input.nextDouble();
        double lat1 = Math.toRadians(x1);
        double lon1 = Math.toRadians(y1);
        double lat2 = Math.toRadians(x2);
        double lon2 = Math.toRadians(y2);
        double distance = radius * Math.acos(Math.sin(lat1) * Math.sin(lat2) + Math.cos(lat1) * Math.cos(lat2) * Math.cos(lon1 - lon2));
        System.out.println("The distance between the two points is " + distance + " km");
    }
}
```

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OUTPUT(Q1):

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shehyar\Desktop\lab 7>javac Q1.java

C:\Users\shehyar\Desktop\lab 7>java Q1
Enter point 1 (latitude and longitude) in degrees: 23.1
124.4
Enter point 2 (latitude and longitude) in degrees: 45.5 22.09
The distance between the two points is 9097.351510107224 km

C:\Users\shehyar\Desktop\lab 7>
```

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QUESTION 2A: Write a program that receives an ASCII code (an integer between 0 and 127) and displays its character

LAB 7

INPUT(Q2A):

Q2a - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q2a
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an ASCII code: ");
        int code = input.nextInt();
        char character = (char) code;
        System.out.println("The character for ASCII code " + code + " is " + character);
    }
}
```

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OUTPUT(Q2A):

 C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.6456]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\shehyar\Desktop\lab 7>javac Q2a.java
```

```
C:\Users\shehyar\Desktop\lab 7>java Q2a
```

```
Enter an ASCII code: 68
```

```
The character for ASCII code 68 is D
```

```
C:\Users\shehyar\Desktop\lab 7>
```


LAB 7

QUESTION 2B: Write a program that receives a character and displays its Unicode

LAB 7

INPUT(Q2B):

 Q2b - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q2b
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch = input.next().charAt(0); // Gets first character of input
        int code = (int) ch;
        System.out.println("The Unicode for the character " + ch + " is " + code);
    }
}
```

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OUTPUT(Q2B):

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shehyar\Desktop\lab 7>javac Q2b.java

C:\Users\shehyar\Desktop\lab 7>java Q2b
Enter a character: a
The Unicode for the character a is 97

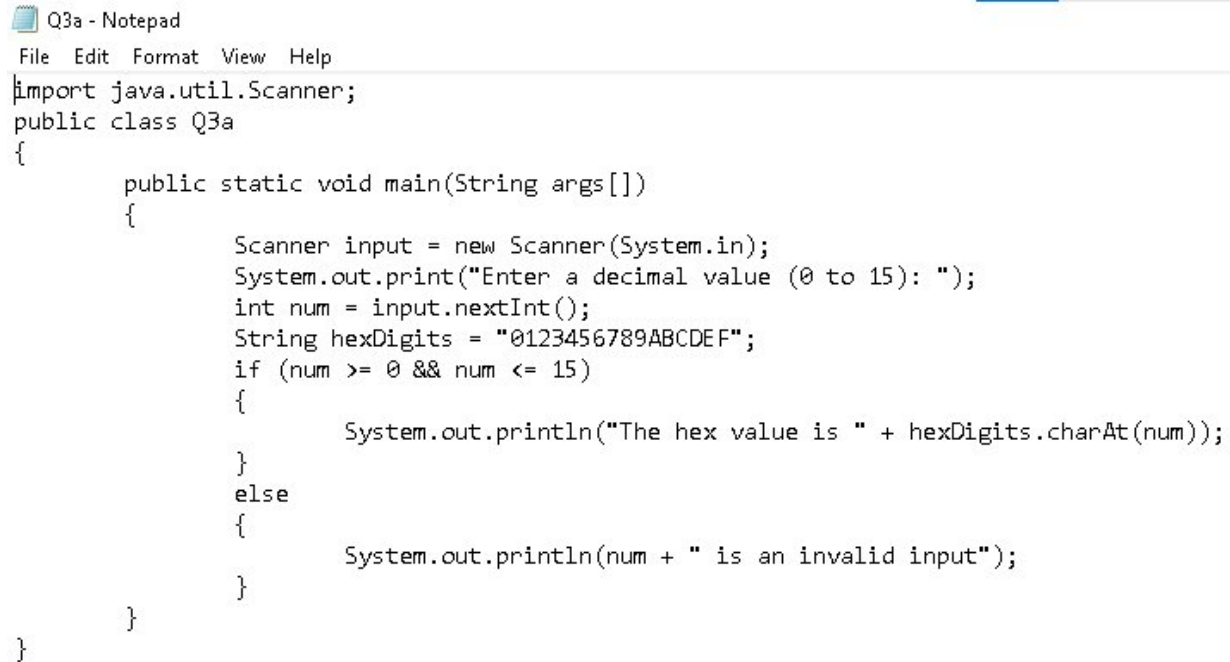
C:\Users\shehyar\Desktop\lab 7>
```

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QUESTION 3A: Write a program that prompts the user to enter an integer between 0 and 15 and displays its corresponding hex number.

LAB 7

INPUT(Q3A):



```
Q3a - Notepad
File Edit Format View Help
import java.util.Scanner;
public class Q3a
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a decimal value (0 to 15): ");
        int num = input.nextInt();
        String hexDigits = "0123456789ABCDEF";
        if (num >= 0 && num <= 15)
        {
            System.out.println("The hex value is " + hexDigits.charAt(num));
        }
        else
        {
            System.out.println(num + " is an invalid input");
        }
    }
}
```

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OUTPUT(Q3A):

```
C:\Users\shehyar\Desktop\lab 7>javac Q3a.java
C:\Users\shehyar\Desktop\lab 7>java Q3a
Enter a decimal value (0 to 15): 12
The hex value is C
C:\Users\shehyar\Desktop\lab 7>_
```

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QUESTION 3B: Write a program that prompts the user to enter a hex digit and displays its corresponding binary number.

LAB 7

INPUT(Q3B):

Q3b - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q3b
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a hex digit: ");
        String s = input.nextLine().toUpperCase();
        String hexDigits = "0123456789ABCDEF";
        if (s.length() != 1 || hexDigits.indexOf(s.charAt(0)) == -1)
        {
            System.out.println(s + " is an invalid input");
            return;
        }
        int num = hexDigits.indexOf(s.charAt(0));
        System.out.println("The binary value is " + Integer.toBinaryString(num));
    }
}
```


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OUTPUT(Q3B):

```
C:\Users\shehyar\Desktop\lab 7>javac Q3b.java  
C:\Users\shehyar\Desktop\lab 7>java Q3b  
Enter a hex digit: A  
The binary value is 1010  
C:\Users\shehyar\Desktop\lab 7>
```

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QUESTION 4: Write a program that displays a random uppercase letter using the `Math.random()` method.

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INPUT(Q4):

 Q4 - Notepad

File Edit Format View Help

```
public class Q4
{
    public static void main(String args[])
    {
        int randomIndex = (int) (Math.random() * 26);
        char randomChar = (char) ('A' + randomIndex);
        System.out.println("Random uppercase letter: " + randomChar);
    }
}
```

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OUTPUT(Q4):

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shehyar\Desktop\lab 7>javac Q4.java

C:\Users\shehyar\Desktop\lab 7>java Q4
Random uppercase letter: V

C:\Users\shehyar\Desktop\lab 7>java Q4
Random uppercase letter: H

C:\Users\shehyar\Desktop\lab 7>
```

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QUESTION 5: Write a program that checks whether a string is a palindrome. A string is a palindrome if it reads the same forward and backward

LAB 7

INPUT(Q5):

 Q5 - Notepad

```
File Edit Format View Help
import java.util.Scanner;
public class Q5
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = input.nextLine();
        String rev = new StringBuilder(str).reverse().toString();
        if (str.equals(rev))
        {
            System.out.println(str + " is a palindrome");
        }
        else
        {
            System.out.println(str + " is not a palindrome");
        }
    }
}
```

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OUTPUT(Q5):

```
C:\Users\shehyar\Desktop\lab 7>javac Q5.java

C:\Users\shehyar\Desktop\lab 7>java Q5
Enter a string: NOON
NOON is a palindrome

C:\Users\shehyar\Desktop\lab 7>java Q5
Enter a string: moon
moon is not a palindrome

C:\Users\shehyar\Desktop\lab 7>
```

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QUESTION 6: Given a string consisting of exactly two words separated by a space. Print a new string with the first and second word positions swapped (the second word is printed first).

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INPUT(Q6):

 Q6 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q6
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter two words");
        String str = input.nextLine();
        int spaceIndex = str.indexOf(" ");
        String firstWord = str.substring(0, spaceIndex);
        String secondWord = str.substring(spaceIndex + 1);
        System.out.println(secondWord + " " + firstWord);
    }
}
```

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OUTPUT(Q6):

```
C:\Users\shehyar\Desktop\lab 7>javac Q6.java
C:\Users\shehyar\Desktop\lab 7>java Q6
Enter two words
Data Science
Science Data
C:\Users\shehyar\Desktop\lab 7>
```

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QUESTION 7: Given a string that may or may not contain a letter of interest. Print the index location of the first and last appearance of f. If the letter f occurs only once, then output its index. If the letter f does not occur, then do not print anything.

LAB 7

INPUT(Q7):

```
File Edit Format View Help
import java.util.Scanner;
public class Q7
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter String ");
        String str = input.nextLine();
        int first = str.indexOf('f');
        int last = str.lastIndexOf('f');
        if (first == -1)
        {
        }
        else if (first == last)
        {
            System.out.println(first);
        }
        else
        {
            System.out.println(first + " " + last);
        }
    }
}
```

LAB 7

OUTPUT(Q7):

```
C:\Users\shehyar\Desktop\lab 7>javac Q7.java
C:\Users\shehyar\Desktop\lab 7>java Q7
Enter String
Office
1 2
C:\Users\shehyar\Desktop\lab 7>_
```

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QUESTION 8: Given a string in which the letter h occurs at least twice. Remove from that string the first and the last occurrence of the letter h, as well as all the characters between them.

LAB 7

INPUT(Q8):

 Q8 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q8
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter A Sentence");
        String str = input.nextLine();
        int first = str.indexOf('h');
        int last = str.lastIndexOf('h');
        String result = str.substring(0, first) + str.substring(last + 1);
        System.out.println(result);
    }
}
```

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OUTPUT(Q8):

```
C:\Users\shehyar\Desktop\lab 7>javac Q8.java
C:\Users\shehyar\Desktop\lab 7>java Q8
Enter A Sentence
In the hole in the ground there lived a hobbit
In tobbit
C:\Users\shehyar\Desktop\lab 7>
```


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QUESTION 9: Given a string. Replace every occurrence of the letter h by the letter H, except for the first and the last ones.

LAB 7

INPUT(Q9):

 Q9 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q9
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter a String");
        String str = input.nextLine();
        int first = str.indexOf('h');
        int last = str.lastIndexOf('h');
        String before = str.substring(0, first + 1);
        String middle = str.substring(first + 1, last).replace('h', 'H');
        String after = str.substring(last);
        System.out.println(before + middle + after);
    }
}
```

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OUTPUT(Q9):

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shehyar\Desktop\lab 7>javac Q9.java

C:\Users\shehyar\Desktop\lab 7>java Q9
Enter a String
In the hole in the ground there lived a hobbit
In the Hole in tHe ground tHere lived a hobbit

C:\Users\shehyar\Desktop\lab 7>_
```

LAB 7

QUESTION 10: You are given a string.

In the first line, print the third character of this string.

In the second line, print the second to last character of this string.

In the third line, print the first five characters of this string.

In the fourth line, print all but the last two characters of this string.

In the fifth line, print all the characters of this string with even indices (remember indexing starts at 0, so

the characters are displayed starting with the first).

In the sixth line, print all the characters of this string with odd indices (i.e. starting with the second

character in the string).

In the seventh line, print all the characters of the string in reverse order.

In the eighth line, print every second character of the string in reverse order, starting from the last one.

In the ninth line, print the length of the given string.

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INPUT(Q10):

Q10 - Notepad

File Edit Format View Help

```
import java.util.Scanner;
public class Q10
{
    public static void main(String args[])
    {
        Scanner input = new Scanner(System.in);
        System.out.println("Enter a Word ");
        String str = input.nextLine();
        System.out.println(str.charAt(2));
        System.out.println(str.charAt(str.length() - 2));
        System.out.println(str.length() < 5 ? str : str.substring(0,5));
        System.out.println(str.substring(0, str.length() - 2));
        String even = "";
        for (int i = 0; i < str.length(); i += 2)
        {
            even = even + str.charAt(i);
        }
        System.out.println(even);
        String odd = "";
        for (int i = 1; i < str.length(); i += 2)
        {
            odd = odd + str.charAt(i);
        }
        System.out.println(odd);
        System.out.println(new StringBuilder(str).reverse());
        String reverseEven = "";
        for (int i = str.length() - 1; i >= 0; i -= 2)
        {
            reverseEven = reverseEven + str.charAt(i);
        }
        System.out.println(reverseEven);
        System.out.println(str.length());
    }
}
```

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OUTPUT(Q10):

```
C:\Users\shehyar\Desktop\lab 7>javac Q10.java
C:\Users\shehyar\Desktop\lab 7>java Q10
Enter a Word
Hello
1
1
Hello
Hel
Hlo
el
olleH
olH
5
C:\Users\shehyar\Desktop\lab 7>_
```